Threat Hunting Methodologies

Threat hunting is a proactive cybersecurity practice that focuses on identifying hidden threats within an organization's environment before they can cause significant damage. In modern Security Operations Centers (SOCs), traditional security tools such as firewalls and intrusion detection systems often rely on predefined rules and signatures, which makes them effective against known threats but less efficient in detecting advanced or unknown attacks. Threat hunting addresses this limitation by enabling analysts to actively search for suspicious activities using hypotheses based on attacker behaviour and known techniques.

One of the key concepts in threat hunting is hypothesis-driven investigation. Instead of waiting for alerts, analysts assume that a potential threat may exist and begin investigating system and network data to validate this assumption. For instance, an analyst may investigate the misuse of valid credentials, such as T1078 (Valid Accounts), by analysing authentication logs for unusual login patterns, privilege escalation, or access from unexpected geographic locations. This approach allows analysts to detect threats that bypass traditional detection mechanisms and remain undetected within the network.

Structured methodologies play an important role in ensuring that threat hunting activities are systematic and effective. Frameworks such as SqRR (Search, Query, Retrieve, Respond) provide a step-by-step approach to identifying and responding to threats. Analysts begin by searching for potential indicators, then query relevant data sources, retrieve meaningful information, and finally respond to any identified threats. Another widely used methodology is TaHiTI (Targeted Hunting integrating Threat Intelligence), which combines threat intelligence with targeted analysis to improve detection accuracy. By using these frameworks, analysts can standardize their hunting processes and improve consistency in their investigations.

Effective threat hunting relies heavily on the analysis of diverse data sources. Organizations generate large volumes of data from endpoints, network devices, and applications, all of which can provide valuable insights into potential threats. Common data sources include Endpoint Detection and Response (EDR) logs, network traffic data such as packet captures and NetFlow, system logs, and external threat intelligence feeds. By correlating information from these sources, analysts can identify patterns and anomalies that may indicate malicious activity. This multi-source analysis is essential for gaining a comprehensive view of the security environment.

The primary objective of threat hunting is to develop the ability to proactively detect threats by analyzing data and identifying patterns associated with attacker behaviour. Analysts must learn how to distinguish between normal and suspicious activities, correlate events across multiple systems, and apply structured methodologies to uncover hidden threats. This requires a combination of analytical skills, technical knowledge, and familiarity with modern security frameworks.

To build expertise in threat hunting methodologies, analysts can study frameworks such as SqRR and TaHiTI through research papers available in the SANS Reading Room. Exploring real-world attack scenarios, such as APT29 operations using the MITRE ATT&CK framework, provides valuable insight into attacker techniques and behaviours. Additionally, practical guidance from platforms like Elastic Security helps analysts understand how to implement

threat hunting techniques in real-world environments. By combining theoretical knowledge with practical exposure, analysts can significantly enhance their ability to detect and respond to advanced threats.